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Milestones Supplemental: Competency-Based Medical Education in U.S. Graduate Medical Education Catalysts and Inhibitors of Implementation

--Manuscript Draft--

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Abstract:	Background Competency-based medical education (CBME) is foundational to U.S. graduate medical education (GME), yet implementation remains challenging. Objective To identify factors that support or inhibit implementation of ACGME Milestones 2.0 within U.S. GME programs. Methods From 2024 to 2025, we conducted an explanatory sequential mixed-methods program evaluation. Phase 1 included 45 semi-structured stakeholder interviews that informed a national Phase 2 survey. This analysis focused on open-ended survey responses, which were thematically analyzed. Results Of 2,567 participants who initiated the survey, 1,306 (51%) completed it. Because dissemination relied on local forwarding, the eligible denominator and response rate could not be determined. Facilitators included a structured developmental framework, shared expectations for progression, support for learner development, and improved assessment precision. Inhibitors included perceived misalignment with clinical performance, inconsistent faculty engagement, implementation and administrative burden, limited tools, grading distortion, and change fatigue. Conclusions These hypothesis-generating findings suggest that Milestones 2.0 can support CBME when integrated into curriculum, assessment, feedback, learner development, and CCC processes. Sustained implementation will require faculty development, practical assessment tools, alignment with clinical performance, and reduced burden.

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Dear Editors,

Please consider our manuscript, **"Competency-Based Medical Education in U.S. Graduate Medical Education: Catalysts and Inhibitors of Implementation,"** for publication in the *Journal of Graduate Medical Education* Milestones Supplement.

This work reports findings from a national program evaluation of ACGME Milestones 2.0, focusing on factors that support or inhibit implementation of competency-based medical education across U.S. graduate medical education programs. Using open-ended responses from a national survey informed by earlier stakeholder interviews, we identified practical facilitators of implementation, including the value of a shared developmental framework, support for learner assessment and development, and improved assessment precision. We also identified important barriers, including perceived misalignment with clinical performance, inconsistent faculty engagement, implementation burden, limited operational tools, grading distortion, and change fatigue.

We believe the manuscript is well suited to this supplement because it moves beyond whether Milestones 2.0 are conceptually aligned with CBME and examines how that framework is experienced in day-to-day implementation. The findings highlight the infrastructure, faculty development, assessment practices, and workflow supports needed to translate a national competency framework into meaningful educational practice.

This manuscript has not been published previously and is not under consideration elsewhere. All authors have reviewed and approved the manuscript. The accompanying survey instrument is included as supplemental material. Because the survey was distributed through program leaders for onward dissemination, a conventional response rate could not be calculated; accordingly, we have framed the survey findings as hypothesis-generating and addressed this limitation in the manuscript.

We appreciate the opportunity to contribute to this special supplement and thank you for your consideration.

Sincerely,

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Milestones Supplemental: Competency-Based Medical Education in U.S. Graduate Medical Education: Catalysts and Inhibitors of Implementation

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Abbreviations: ACGME, Accreditation Council for Graduate Medical Education; CBME, competency-based medical education; CCC, Clinical Competency Committee; EPA, Entrustable Professional Activity; GME, graduate medical education; ILP, individualized learning plan.

Abstract

Background: Competency-based medical education (CBME) is foundational to U.S. graduate medical education (GME), yet implementation remains challenging.

Objective: To identify factors that support or inhibit implementation of ACGME Milestones 2.0 within U.S. GME programs.

Methods: From 2024 to 2025, we conducted an explanatory sequential mixed-methods program evaluation. Phase 1 included 45 semi-structured stakeholder interviews that informed a national Phase 2 survey. This analysis focused on open-ended survey responses, which were thematically analyzed.

Results: Of 2,567 participants who initiated the survey, 1,306 (51%) completed it. Because dissemination relied on local forwarding, the eligible denominator and response rate could not be determined. Facilitators included a structured developmental framework, shared expectations for progression, support for learner development, and improved assessment precision. Inhibitors included perceived misalignment with clinical performance, inconsistent faculty engagement, implementation and administrative burden, limited tools, grading distortion, and change fatigue.

Conclusions: These hypothesis-generating findings suggest that Milestones 2.0 can support CBME when integrated into curriculum, assessment, feedback, learner development, and CCC processes. Sustained implementation will require faculty development, practical assessment tools, alignment with clinical performance, and reduced burden.

Introduction

Competency-based medical education (CBME) is not new; the concept traces its origin to a 1978 World Health Organization manuscript in which the authors described the goal of a medical education program as being defined by the outcomes expected of graduates: "The intended output of a competency-based program is a health professional who can practice medicine at a defined level of proficiency, in accord with local conditions, to meet local needs."[1] It would take 20 years for graduate medical education (GME) in the United States to move toward implementation of CBME with the introduction of the ACGME core competencies in 1999. In 2012, these competencies underwent further development to better define them, mitigate ambiguity, and align them with CBME principles through subcompetencies and Milestones.[2] There is no uniform methodology for implementing CBME. The approach seeks to make explicit the developmental progression of medical learners while considering the needs of patients within local contexts.[3] Additional fundamental characteristics include a shift from time- and experience-based metrics toward authentic outcomes and learner-centered program design.[4]

The use of competencies and Milestones became an accreditation requirement for all U.S. GME programs with phased implementation of the Next Accreditation System beginning in 2013. While every program reports on learner progression, little is known about the local, regional, or national characteristics associated with successful CBME implementation. Much of the implementation literature comes from systems outside the United States, including Canada's CanMEDS/Competence by Design program. A narrative review identified broad stakeholder engagement and leadership, faculty and resident development, and systems to support curricular adoption as important implementation themes.[5]

As part of a national program evaluation of CBME in the U.S. GME context, we aimed to identify factors that supported (catalysts) or inhibited (inhibitors) successful implementation of CBME through ACGME Milestones 2.0.

Methods

In July 2024, the ACGME recruited a team of GME educational leaders to review the impact and effectiveness of Milestones 2.0 using a principles-focused program evaluation approach to develop the guiding evaluation questions. The evaluation used a two-phase explanatory sequential mixed-methods design, with Phase 1 findings informing the Phase 2 survey. Phase 1 used inductive qualitative methods, including semi-structured interviews lasting 40 to 60 minutes with purposively sampled program directors, CCC chairs, and assistant program directors. The sample was selected for diversity in program type, region, and institutional context and included both former Milestones Development Group members and nonmembers. Interviews were recorded, transcribed, and thematically analyzed.

Phase 2 expanded the sample to faculty, trainees, CCC members, and administrative staff across ACGME-accredited residency and fellowship programs. The national survey was distributed through standard ACGME communication channels to program leaders for dissemination within their programs. Survey domains included perceptions and use of Milestones, CCC and data processes, implementation experience, and respondent demographics. The full survey instrument is provided as online supplementary data. Because dissemination relied on local forwarding, the number of individuals who received or were eligible to complete the survey could not be determined.

This manuscript focuses on the open-ended Phase 2 responses concerning facilitators and barriers to implementation. Qualitative responses were thematically analyzed. ChatGPT

(OpenAI, GPT-5.6 Sol, accessed February 6, 2026) was used to generate an initial thematic organization of the open-ended survey data. Two authors (R.R.C. and S.A.M.) independently cross-checked the AI-generated themes against source data for accuracy, with a third author serving as a tiebreaker for disagreements.

The University of Illinois Chicago Institutional Review Board determined that this program evaluation did not constitute research involving human subjects as defined by federal guidelines.

Results

A total of 2,567 participants initiated the survey, of whom 1,306 (51%) completed it. Because the survey was disseminated through program leaders for onward distribution, the number of eligible recipients could not be determined and a conventional response rate could not be calculated. Median completion time was 10 minutes. Open-ended responses included 791 comments related to facilitators of implementation and 1,749 comments related to barriers to implementation.

Qualitative analysis identified 3 broad themes that supported CBME implementation and 7 themes that inhibited successful implementation.

Facilitators of Adoption and Implementation

Analysis identified three major themes describing factors that facilitated adoption and implementation of Milestones 2.0: a comprehensive and structured framework, support for meaningful assessment and learner development, and improved assessment precision with Milestones 2.0 (Table 1). The first two themes accounted for most facilitator-related responses, while comments specifically addressing improvements associated with Milestones 2.0 were less frequent.

Comprehensive, Structured Framework

The most prominent facilitator theme was the value of the Milestones as a structured developmental framework. Respondents described the Milestones as providing a common language, clearer expectations for progression, and a shared framework through which faculty could conceptualize resident performance. One respondent described them simply as providing a "common language and framework," while another explained that the Milestones "are a clear way to understand what is expected of residents as they progress through their program and they help the faculty create a shared mental model for evaluating residents."

Responses also suggested that the framework influenced educational design beyond individual learner assessment. Participants described the Milestones as "guideposts to make sure our curriculum is addressing the milestones for residents" and noted that "there is good specificity to help drive curriculum." Thus, implementation was described as moving from assigning milestone ratings to using the framework to establish expectations, align curriculum, and create a shared understanding of competency across faculty and learners.

Support for Meaningful Assessment and Learner Development

The second heavily represented theme concerned the ability of the Milestones to organize assessment around developmental progression and support learner growth. Respondents described their use in direct observation, feedback, individualized learning plans, and identification of curricular and/or learner gaps. One participant noted that standardization was helpful in "focusing faculty on measuring against a shared mental model, and refrain from comparing fellow performance against each other, and to focus on the individual learning trajectory." Another described the value of "concrete language that is behavioral,

developmental..., and using it as a shared mental model and tool for conversation and individualized learning plan development."

Several responses illustrated what implementation looked like operationally. Participants described using Milestone data for "individualized learning plans, knowledge assessment, curriculum development" and for "identifying gaps in the curriculum or between learners." Others emphasized direct observation and multisource feedback. Collectively, these responses suggest that the Milestones facilitated CBME when programs used them as an organizing framework for longitudinal assessment, feedback, individualized development, and curricular improvement, rather than solely as a semiannual reporting requirement.

Improved Assessment Precision With Milestones 2.0

A smaller but distinct group of responses specifically identified Milestones 2.0 as an improvement over the prior framework. Participants attributed this to clearer language, more clinically relevant content, and more understandable developmental progression. One respondent reported that "the simplification of language was the main improvement" and that the addition of clinically relevant items allowed the program "to tie our evaluation process more closely to milestones to increase accurate assessment and reporting." Another noted that "reading left to right across a row, Milestones 2.0 has a clear progression of skills."

For some programs, these improvements supported substantial integration into the educational program. One respondent stated, "Milestones 2.0 tool is extremely effective as a whole. We have built [our] mid-year assessment period; final assessment period; and rotation goals and objectives around this tool." These examples demonstrate a deeper level of implementation in which Milestones 2.0 became embedded within the program's assessment inventory and curricular expectations.

Threats to Adoption and Implementation

Seven major themes represented threats to adoption and implementation: misalignment with real-world competence, inconsistent application and faculty buy-in, time burden, lack of templates and implementation tools, administrative and reporting burden, grading distortion, and change fatigue associated with the transition to Milestones 2.0. Misalignment with real-world competence was by far the most frequently represented theme, followed by inconsistent application/faculty buy-in and time burden. Lack of tools and administrative burden were also recurrent concerns, whereas grading distortion and transition-related change fatigue were less frequently identified. (Table 2)

Misalignment With Real-World Competence

The dominant threat concerned the perceived disconnect between Milestone constructs and authentic clinical performance. Respondents questioned whether some Milestones represented actual competence, whether developmental sequences reflected how residents acquire skills, and whether the framework could capture readiness for independent practice. Some comments were unequivocal, including “The[y] do not reflect real world competency.”

Other respondents described developmental incongruities, noting that learners could demonstrate higher-level behaviors without necessarily demonstrating behaviors assigned to earlier levels. This perceived disconnect was particularly apparent for competencies that were difficult to observe directly or whose applicability varied by specialty. Taken together, these responses suggest that adoption is threatened when faculty cannot readily connect Milestone language and levels to the clinical behaviors they actually observe and value.

Inconsistent Application and Faculty Buy-In

A second prominent threat was variable faculty understanding, interpretation, and engagement. Respondents reported difficulty getting faculty to understand and appropriately apply the framework. One participant stated, “It's hard to get all of the faculty to understand how they work and fill out assessments appropriately. Deciding where a resident stands on milestones is somewhat subjective.”

More fundamentally, some faculty did not find the scale intuitive. One respondent explained that Milestones “have never felt like a ‘natural’ assessment scale” and that because faculty did not find them meaningful, “they continue to assign scores based on subjective sense of how a resident is doing rather than use the anchors.” These comments suggest a reciprocal threat: inconsistent understanding reduces confidence in Milestone ratings, while lack of confidence in the framework may further diminish faculty engagement.

Time Burden

Time was another highly recurrent implementation threat. Respondents described the cumulative time required to develop assessments, collect meaningful data, conduct CCC deliberations, develop ILPs, and translate assessment information into Milestone ratings. Comments ranged from the concise, “The discussions are endless. Extremely time consuming!” to detailed descriptions of the workload associated with implementation.

One respondent summarized the challenge as requiring “TIME and buy in from others working with residents to get meaningful feedback to apply to milestones” while another described the creation of supporting documents as “tedious and time consuming.” These responses indicate that time burden was not restricted to reporting; it permeated the assessment system required to generate credible Milestone judgments.

Lack of Templates and Implementation Tools

Respondents also frequently identified the absence of standardized, practical tools for translating Milestones into frontline assessment. Programs reported having to create their own evaluation systems, map Milestones to clinical experiences, and determine how to assess difficult competencies. One respondent characterized the system as "too complex" with "no standardized tools from the ACGME on how to actually assess," resulting in each program needing to develop its own assessment system.

Requests included specialty-specific tools, standardized evaluation forms, templates, and better approaches to translating workplace assessment into Milestone judgments. Several respondents also requested or made note of Entrustable Professional Activities (EPAs). This suggests a misunderstanding of the structure and function of EPAs versus competencies. This theme suggests that the availability of a national competency framework alone may be insufficient for implementation without accompanying operational tools even with the existence of specialty-specific implementation guides.

Administrative and Reporting Burden

Related to, but distinct from, time burden was the administrative complexity created by assessment aggregation and reporting requirements. One program described shortening rotation evaluations by matching specific Milestones to particular rotations, but noted that doing so made "the back end of aggregating the feedback for the CCC and milestones reporting more burdensome."

Another respondent specifically described "the burden of reporting to ACGME" and warned that duplicative reporting requirements could "lead to less meaningful data and PD burnout." Thus, efforts to make Milestone assessment more feasible at the frontline could shift rather than

eliminate workload, creating additional complexity for CCCs, program leadership, and administrators.

Grading Distortion and Inflation

Although less frequently identified, respondents described misinterpretation of Milestone levels as conventional grades. One participant explained, “There's lots of grade inflation because many people equate a level of 3 with a grade of C or something like that. We just can't shake that idea from people's heads.” Another described a “real sense of ‘grade inflation’ based on years in the program or whether or not the trainees are pleasant people.”

These responses highlight a threat to the developmental intent of CBME: when Milestone levels are interpreted as normative grades rather than a developmental progression, ratings may reflect expectations based on postgraduate year, interpersonal impressions, or reluctance to assign apparently “low” scores.

Change Fatigue Associated With Milestones 2.0

Finally, a small number of responses described challenges specifically associated with transitioning from the original Milestones to Milestones 2.0. One respondent described the transition as “an adjustment” and reported counseling fellows that changes in questions and scoring could produce apparent decreases that did not indicate regression in learning.” Other responses emphasized the need for faculty training to understand and apply changes consistently. Although this theme was substantially less frequent than concerns about validity, faculty engagement, or workload, it illustrates the implementation costs associated with revision of a national assessment framework.

Discussion

Across stakeholder groups, Milestones 2.0 were viewed as conceptually aligned with the principles of CBME. Programs described meaningful integration of the framework, including curriculum mapping, assessment redesign, longitudinal review of performance, and structured CCC processes. However, adoption remained uneven. Implementation requires substantial faculty time, infrastructure, and cultural alignment. Without sufficient support and visible educational return on investment, CBME risks being perceived as an accreditation requirement rather than as an educational transformation supporting learner readiness for independent practice.

Adoption is enabled when Milestones function as a shared developmental framework embedded within assessment, curriculum, feedback, and learner development systems. Conversely, adoption is threatened when workload burden, inconsistent faculty understanding and engagement, and perceived misalignment between Milestone constructs and authentic clinical performance erode confidence in their educational value.

What Implementation of CBME Actually Looks Like

The data suggest that successful implementation of Milestones-based CBME requires a different way of working. Programs described redesigning evaluation tools to align assessment with Milestone subcompetencies, structuring CCC review around longitudinal performance data, and using Milestone information to inform individualized learning plans and remediation. Meaningful implementation also appears to depend on direct observation, workplace-based assessment, multisource feedback, and high-quality narrative data capable of informing judgments about learner progression.[6]

Programs also described the need for documentation and data systems capable of capturing competency progression over time. Some programs are using curriculum mapping to identify where competencies are taught and assessed, reveal gaps in curriculum and assessment, and support ongoing improvement. Implementation also requires faculty and program leadership time, sustained engagement across the assessment process, and institutional and administrative infrastructure capable of supporting longitudinal competency assessment.[7,8] Structured competence committee processes are similarly central to translating longitudinal assessment data into progression decisions.[9]

Recommendations for Strengthening Adoption and Implementation

Based on these findings, we recommend three practices to enhance successful implementation:

- Strengthen faculty development and shared understanding.
- Improve alignment between Milestones, assessment, and clinical practice.
- Reduce implementation burden and support sustainability.

Faculty development should focus on shared mental models of competency progression and Milestone rating scales, as well as direct observation and workplace-based assessment.[10] Additional attention is warranted for competencies respondents identified as particularly difficult to observe and assess, including professionalism and systems-based practice. Programs, institutions, specialty organizations, and accrediting bodies should also support implementation with user-friendly assessment tools, such as templates and specific behavioral anchors, rather than requiring individual programs to construct these systems independently.

The most prominent threat identified in our analysis was difficulty translating Milestone language and developmental levels into authentic clinical performance. Respondents described some Milestones as difficult to observe, thus not reflective of real-world competence. Enhancing

alignment with clinical practice may therefore require specialty-specific behavioral anchors, practical examples of assessment strategies, and clearer links between Milestone levels and observable workplace performance. Programs and specialty organizations should also consider how Milestones can be more deliberately harmonized with Entrustable Professional Activities and other competency frameworks to reduce duplication and improve translation of workplace performance.

Demonstrating relationships between Milestone implementation and meaningful educational, clinical, or patient-centered outcomes may also be important for increasing perceived value and faculty investment. Greater transparency regarding these outcomes could help programs demonstrate return on the substantial time and resources required for implementation.

Finally, successful implementation requires explicit attention to sustainability. Time and administrative burden were recurring threats to adoption. Programs should explore streamlined assessment approaches that prioritize brief, authentic workplace observations and meaningful narrative data. At the system level, efforts to reduce redundant reporting, provide usable implementation tools, and integrate assessment into existing clinical workflows may help preserve the educational intent of the Milestones while reducing implementation burden.

Overall, the data suggest that Milestones 2.0 provide a conceptually aligned framework for CBME implementation. Programs that integrate Milestones into curriculum design, assessment systems, learner development, and CCC processes demonstrate meaningful adoption. However, sustained implementation depends on institutional capacity, faculty engagement, assessment approaches that reflect authentic clinical practice, and an implementation burden proportionate to the educational value of the framework.

Limitations

This evaluation has several limitations. Because the survey was disseminated through intermediary program leaders, the number of eligible recipients was unknown, precluding calculation of a conventional response rate and limiting assessment of respondent representativeness. Open-ended responses were optional and may overrepresent participants with stronger views or greater engagement with Milestones 2.0. Comment frequencies should therefore not be interpreted as prevalence estimates. These findings are best considered hypothesis-generating and may not capture the full range of implementation experiences across all programs and specialties.

Conclusions

CBME aims to ensure that trainees graduate ready for unsupervised practice. These findings suggest that Milestones 2.0 can contribute to that goal. Respondents described meaningful implementation, including curriculum remapping, assessment redesign, and structured Clinical Competency Committee processes, while also identifying substantial implementation challenges. Milestones appear most useful when they function as a shared developmental framework embedded within curriculum, assessment, feedback, coaching, and progression decisions. Their educational value may be diminished when workload burden, faculty skepticism, and assessment ambiguity erode confidence in the framework.

The themes identified by respondents closely mirror the core components framework described by Van Melle and colleagues, including an organizing framework of competencies, sequenced progression of learning experiences, learning and teaching tailored to competencies, and programmatic assessment.[6] This provides an example of CBME theory becoming visible in the

day-to-day work of programs implementing Milestones 2.0 across diverse clinical learning environments.

Milestones 2.0 should therefore be understood as part of an evolving process rather than a finished solution. Continued progress will require investment in faculty development, assessment quality, curricular alignment, and meaningful use of Milestone data to support learner growth.

Acknowledgments

The authors thank Laura Edgar, PhD, for her leadership and support of this work.

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Table
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Table 1. Major Themes Facilitating Adoption and Implementation of Milestones 2.0

Major theme	Exemplar quotes
Comprehensive, Structured Framework	"Being able to sit down and visually see where they resident is as at that moment in time to determine if they are needing more training in a specific area or if they are on par/exceeding expectations." "Specific and timely ratings. A good ordinal scale to show improvement or stagnation over time." "the structure and steps down/ladder effect works great to keep everyone on the same page. It gives everyone specific examples of where progress sits."
Supports Meaningful Assessment and Development	"Focus on Meaningful Assessment. The framework emphasizes direct observation and real-world application, helping programs assess competencies in authentic clinical settings. Encourages multi-source feedback, promoting holistic evaluations rather than relying solely on faculty assessments." "Some milestones that are more likely to be evaluated objectively have supported efforts in learner assessment and feedback." "Patient care and medical knowledge are most effective, followed by ICS. They are helpful for comparing residents within classes, across the residency program, and across programs."
Improved Assessment Precision	"The Milestones 2.0 helps the program gauge trainee development because of their straightforward nature, which leads to a more objective assessment." "Milestone 2.0 are an improvement as a tool for overall assessment compared to the previous version. It allows better assessment of concepts rather than specific disease entities or procedures." "The language is easier to understand in milestones 2.0. The examples are also very useful."

Table
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Table 2. Major Themes Threatening Adoption and Implementation of Milestones 2.0

Major theme	Additional exemplar quotes
Misalignment with Real-World Competence	"...indistinct differentiation within and between levels, they don't flow like clinicians think and they aren't intuitive to use when assessing residents." "...there are too many milestones that are reasonable attributes for a physician to have, but unwieldy to consistently assess." "can be hit or miss (based on the patients seen during a rotation) whether there is an opportunity to assess the fellow on all items in each milestone category"
Inconsistent Application / Faculty Buy-In	"My biggest challenge is faculty compliance. I educate them annually on the milestones and how to apply them, and the instructions are on every evaluation. but that's an issue I'm working with internally." "getting all the faculty to be on same wavelength" "getting all faculty outside of core to understand the language around milestones"
Time Burden	"The process is very time-consuming and not always clearly applicable to evaluating and supporting the progress of all learners." "they take too long to fill out and make the CCC meetings too long if CCC members don't take the time fill them out before hand." "Time, paperwork, no help with the additional work, not a helpful tool."
Lack of Templates / Tools	"Could Milestones be competency-based (i.e., EPAs)? An entrustability scale may be more meaningful to CCCs and the trainees themselves." "Implementation of EPAs to feed into Milestone assessment." "If they are shorter/ more concise. Less Milestones. Maybe change to Entrustable Professional Activities (EPAs)"
Administrative / Reporting Burden	"Updating evaluation forms to collect meaningful assessment." "Tracking along the curriculum where, explicitly each milestone is being assessed." "The complexity of mapping to rotation evaluations. Faculty concerned about the length of these rotation evaluations."

Major theme	**Additional exemplar quotes**
Grading Distortion / Inflation Concerns	"People grade inflating. I always have to talk people lower in CCC" "Very good residents are often 'graded down' in an effort to advance them slowly throughout their residency. But it works best for struggling residents"
Change Fatigue	"time spent in re-creating evaluation forms to be updated for Milestones 2.0" "The change from 1.0 to 2.0 required a complete revamping of our evaluation systems. This was a lot of work, but the revised system is better."

Supplemental Material
Click here to access/download;Supplemental Material;Supplement-FINAL Milestones Survey Jan 2025.docx

Milestones 3.0 Survey

Introduction

Thank you for taking the time to provide your experience with Milestones 2.0. The Milestones 2.0 development process concluded in 2023. Efforts by the ACGME then moved to conducting a principles-focused program evaluation in collaboration with Dr. Laura E. Hirshfield, PhD, at University of Illinois Chicago (UIC). Dr. Hirshfield has developed a survey guided by the principles upon which the Milestones were first developed and interviews conducted with 45 stakeholders. The survey is designed to elicit your perspectives and experiences regarding the implementation and use of Milestones 2.0. Only aggregated summary results will be shared with the ACGME.

As a user of the Milestones, we invite you to participate in helping the ACGME identify and understand how the Milestones are, or are not, working within your context. By keeping the survey anonymous, we strongly encourage you to provide your honest feedback to improve the Milestones framework to better serve the GME community and its learners.

You may not be familiar with one phrase used within the survey – harmonized Milestones. Harmonized Milestones include a set of 11 subcompetencies across Interpersonal and Communication Skills, Practice-Based Learning and Improvement, Professionalism, and Systems-Based Practice that describe knowledge, skills, and behaviors applicable across all surgical and medical specialties. Each specialty was offered the opportunity to make specialty-specific edits. Few specialties made substantial changes to the harmonized Milestones.

The UIC Institutional Review Board determined that this evaluation does not constitute research involving human subjects as defined by federal guidelines.

The survey should take approximately 15 minutes to complete.

Section 1: Demographics

1. What is your primary role in GME? (check one)
    a. Program director
    b. Associate/assistant program director
    c. Physician faculty
    d. Non-physician faculty
    e. Designated institutional official (DIO) *(If selected, skip Q3, 5,7, 8)*
    f. Resident/fellow
    g. Non-physician administration/leadership (e.g., coordinator, nurse)
    h. Other (please specify):_____
2. How many years have you been in your current role?
3. For the purposes of this survey, please focus on one specific specialty/subspecialty for your responses. What residency or fellowship program are you responding for? (check one/drop down)
4. Which of the following professional degrees have you earned? (Check all that apply)
5. If applicable, approximately how many of each the following are in your program?
    a. Residents:

b. Fellows:
c. Preliminary residents:
Please check here if the above are estimates ☐

6. Which of the following best describes your program? (check all that apply)
    a. Academic (i.e. hospitals that are affiliated with medical schools and focus on education and research)
    b. Community (i.e., non-academic hospitals that provide a range of services to local communities)
    c. Rural
    d. Federally Qualified Health Center (FQHC)
    e. Veteran Affairs (VA)
    f. Uncertain
    g. Other (please specify): _____
7. What is/was your role(s) with your program's Clinical Competency Committee (CCC)? (select any that apply, past or present)
    a. Member
    b. Chair
    c. Non-voting member
    d. Chief resident member
    e. Not a member
8. Please specify (EXCLUDE IF SELECT 7e)
    a) What other specialty(ies) or subspecialty(ies) does your CCC review?
    b) How many members serve on your program's CCC (including yourself)?
    c) How many non-physician members serve on the CCC?
    d) How many times per year does your CCC meet?
    e) Does your CCC receive ongoing faculty development? y/n

Section 2: Opinions

For the following questions, please only answer based on the residency or fellowship you selected in the previous section. For DIOs, please answer based on your overall experience across specialties.

1. How important is it that all learners of the same specialty are assessed against the same benchmarks outlined in the Milestones?

    Not at all important | Somewhat important | Important | Very important | Extremely important

    If you selected "not at all important" or "somewhat important," please elaborate:

2. How satisfied are you with Milestones 2.0 as a tool for supporting the adoption of competency-based medical education (CBME) components (assessment, curriculum, feedback, programmatic assessment, individualized learning plans, etc.) for each of the following?
    a. All six Core Competencies
    b. Patient Care (PC)
    c. Medical Knowledge (MK)
    d. Systems-Based Practice (SBP)
    e. Practice-Based Learning and Improvement (PBLI)
    f. Professionalism (PROF)
    g. Interpersonal and Communication Skills (ICS)

    Very unsatisfied | Unsatisfied | Satisfied | Very satisfied | Unsure

    If you selected "very unsatisfied" or "unsatisfied," please elaborate:____

3. How effectively do Milestones 2.0 support each of the following?
    a. Program improvement goals (e.g., increased participation in quality improvement activities)
    b. Educational outcomes for learners (e.g., higher in-training exam scores)
    c. Learner self-assessment (i.e., resident/fellow may perform informed self-assessment)
    d. Patient outcomes
    e. Individualized learning plans (e.g., customized set of activities to achieve personalized educational goals)

    Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

    If you selected "not at all effectively" or "somewhat effectively," please elaborate:____

4. How satisfied are you with the following aspects of Milestones 2.0?
    a. Simple language and design.
    b. Distinction between meaningful levels of competency (e.g., between Milestone level 2 and 3)
    c. Representation of realistic progression of knowledge, skills, and behaviors (i.e., across the 5 levels)
    d. Supplemental Guide (ACGME companion document to Milestones 2.0)
    e. Harmonized Milestones that were developed for all specialties
    f. Representative of national standards

    Very unsatisfied | Unsatisfied | Satisfied | Very satisfied | Unsure

Please elaborate, as desired:

5. How effectively do harmonized Milestones foster consistent curriculum and assessment across each of the following?

    a. Institutions

    b. Departments
    c. Specialty societies/boards

Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

Please elaborate, as desired:

6. How effectively do the content and competencies assessed by Milestones 2.0 measure the knowledge, skills, and behaviors required by learners to serve the public without supervision?
   Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

    Please elaborate, as desired:

7. How effectively did Milestones 2.0 promote program improvement of the following?

    a. Assessment of learners

    a. Feedback to learners
    b. Individualized learning plans
    c. Curriculum for harmonized Milestones (PROF, PBLI, ICS, and SBP)

Not at all effectively | Somewhat effectively | Effectively | Very effectively | Unsure

Please elaborate, as desired:

8. How satisfied are you with the following aspects of Milestones reporting:

    a. Frequency (i.e., reporting Milestones twice per year)

    b. Timing (i.e., reporting Milestones October-January and April-June)

Very unsatisfied | Unsatisfied | Satisfied | Very Satisfied | Unsure

Please elaborate, as desired:

Section 3: Open-Ended Feedback

9. What aspects of Milestones 2.0 are most effective in supporting assessment and learner development? (Open text response)

10. What challenges have you encountered in implementing/ongoing use of Milestones 2.0? (Open text response)

11. What additional tools or resources would help you better use Milestones 2.0? (Open text response)

12. What, if any, faculty development is offered at your program regarding Milestones and/or assessment?

13. Do you have any additional feedback or suggestions for improving the Milestones? (Open text response)

14. Are you willing to take an additional 10-15 minutes to answer a few more in-depth questions regarding the milestones? Y/N (*yes go to section 4, no end survey*)

    If Y, select the type of questions you want to answer (check all that apply)

    a. Data use (two questions)
    b. Implementation and assessment (five questions)
    c. Clinical Competency Committee (CCC) (five questions)

Section 4: Additional questions for those who selected 14Y

Data Use

1. Do ___ use their Milestones data? Y/N/unsure matrix
   a. Learners
   b. Faculty members (core)
   c. Faculty members (frontline)
   If yes, please elaborate as desired:

2. Does your Program Evaluation Committee use Milestones data? If so, how? (Y/N/Unsure) Explain:

Implementation and Assessment

3. How are Milestones 2.0 used to identify and support the learner who is struggling?
4. How are Milestones 2.0 used to identify and support the accelerated learner?
5. On average, how many assessments per resident per competency are performed every six months?
6. Do the Milestones support equitable evaluation of all learners? If so, how or how not? (Y/N) Explain:
7. What does effective implementation of Milestones look like?

Clinical Competency Committee (CCC) Questions:

1. Describe the processes used by the CCC to assign milestones semi-annually, including data sources.
2. What is the role of your program director on the CCC?
   a. CCC chair

    b. CCC member
    c. CCC observer
    d. Program director does not attend
3. Do you have a patient/public representative on your CCC? y/n
    a. If yes, describe their role and how they participate_
4. What percentage of residents/fellows does your CCC review at each meeting?
    a. Less than 25 percent
    b. 26-50 percent
    c. 51-75 percent
    d. 76-100 percent
    e. Unsure
5. Who provides feedback from the CCC to the resident/fellow?
    a. Program director
    b. Assistant/associate program director
    c. CCC member (not program director or assistant/associate program director)
    d. Resident/fellow advisor